Dynamics of local quantum information in random unitary circuits

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Information Lattice

Local information[1][2]

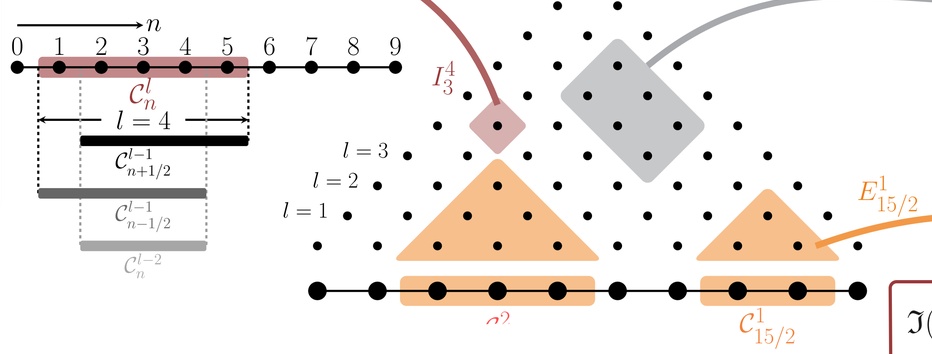
$I^l_n = E^l_n - E^{l-1}_{n-1/2} - E^{l-1}_{n+1/2} + E^{l-2}_n$

"Information in a subsystem that is not contained in any of its constituents"

$0 \le I^l_n \le 2$

Continuum

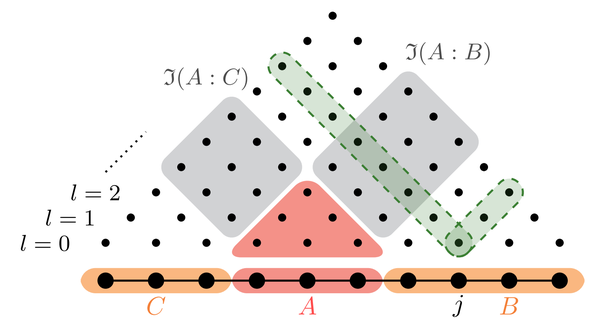
$$\mathtt{I}(\mathtt{n},\mathtt{l}) = \left(\partial^2_{\mathtt{l}} - \frac{1}{4}\partial^2_{\mathtt{n}}\right)\mathtt{E}(\mathtt{n},\mathtt{l})$$



$\mathfrak{I}(\mathcal{C}^2_3 : \mathcal{C}^1_{15/2} | \mathcal{C}^1_{11/2})$
Conditional mutual information

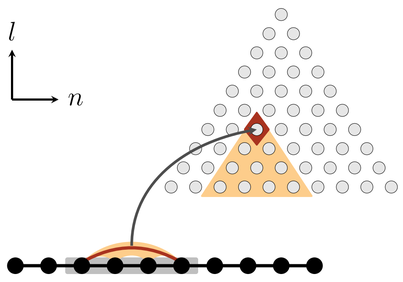
$E_A = L_A - S_A$
von Neumann information

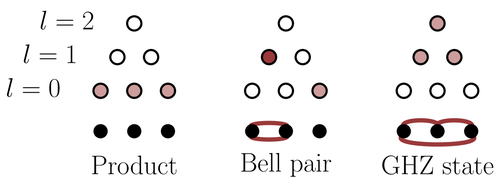
Constraints



$\mathfrak{I}(A:B) + \mathfrak{I}(A:C) = 2L_A - 2E_A$ is required by Purity of the state.

$$2I^0_j + \left(\sum_{l=1}^{L-j} I^l_{j+l/2} + \sum_{l=1}^{j} I^l_{j-l/2}\right) = 2 \quad \forall j.$$





Additivity:

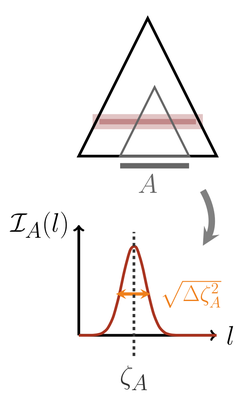
- $\sum_{n,l} I^l_n = L$
- $E_A = \sum_{(n,l)\in A} I^l_n$

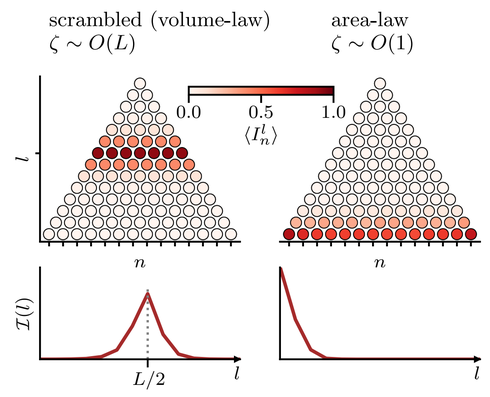
Diagnostics

$$\mathcal{I}_A(l) = \sum_{n:(n,l)\in A} I^l_n$$

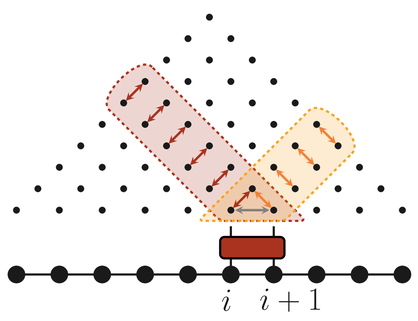
$$\rightarrow \quad \zeta_A = \frac{1}{E_A}\sum_l l\, \mathcal{I}_A(l)$$

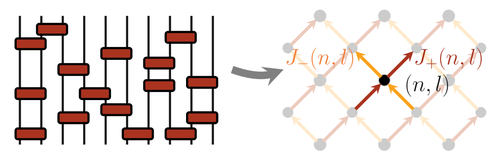
$$\rightarrow \quad \Delta\zeta^2_A = \frac{1}{E_A}\sum_l l^2\, \mathcal{I}_A(l) - \zeta^2_A$$





Information currents in local circuits





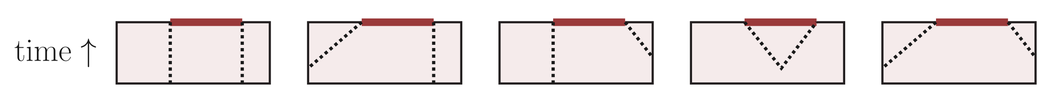
"local flow of a globally conserved quantity in the bulk of the information lattice"

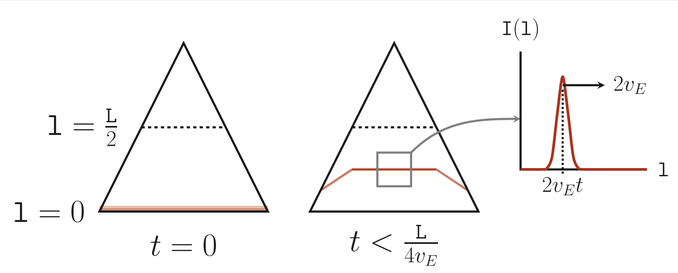
$$\partial_t \mathtt{I}(\mathtt{n},\mathtt{l},t) = -\left(\partial_{\mathtt{l}} + \frac{1}{2}\partial_{\mathtt{n}}\right) J_+(\mathtt{n},\mathtt{l},t) - \left(\partial_{\mathtt{l}} - \frac{1}{2}\partial_{\mathtt{n}}\right) J_-(\mathtt{n},\mathtt{l},t)$$

Local information dynamics at the leading order

Leading order entanglement dynamics from entanglement membrane [3]

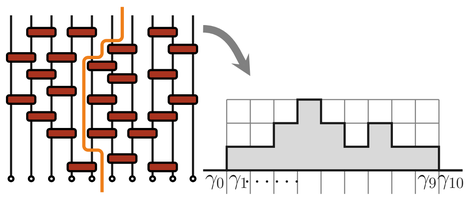
$$S(\mathtt{n},\mathtt{l},t) = \min\left\{2v_E t,\ v_E t + \left(\mathtt{n} - \frac{\mathtt{l}}{2}\right),\ v_E t + \left(\mathtt{L} - \mathtt{n} - \frac{\mathtt{l}}{2}\right),\ \mathtt{l},\ (\mathtt{L} - \mathtt{l})\right\}$$

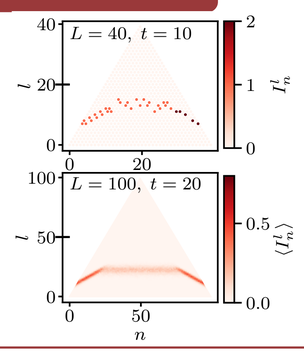




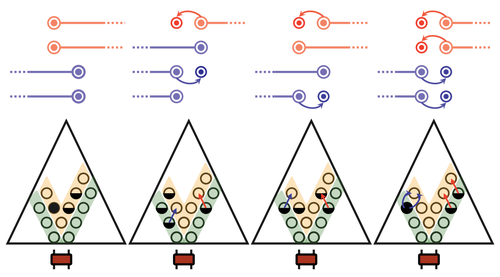
Haar random circuits in the $q \to \infty$ limit

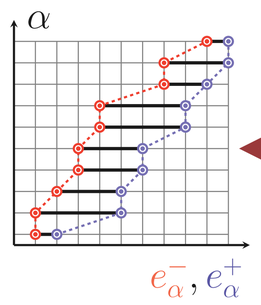
- Entanglement computed by a minimal cut bound.
- Dynamics described by a KPZ height field.

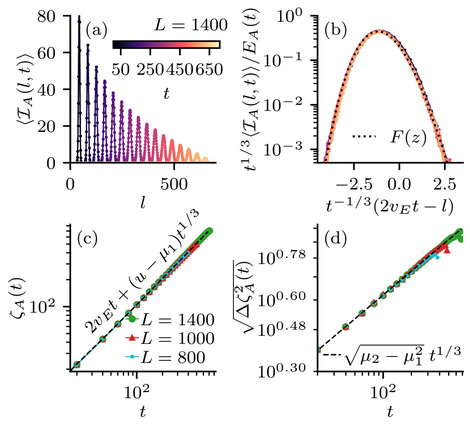




Classical process







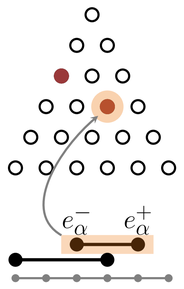
Bigram representation

L one-dimensional objects,

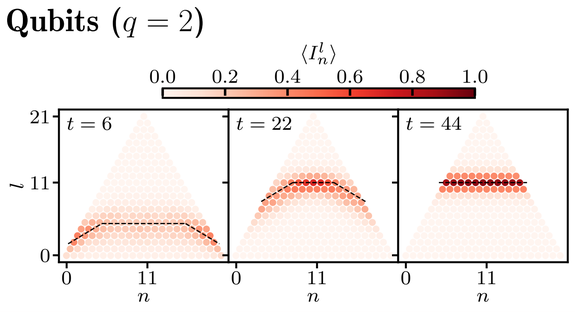
$$\mathcal{B} = \left\{(e^-_0, e^+_0), (e^-_1, e^+_1), ..., (e^-_{L-1}, e^+_{L-1})\right\}$$

$$I^l_n = \sum_{\alpha=0}^{L-1} \delta(e^-_\alpha, n - l/2)\ \delta(e^+_\alpha, n + l/2)$$

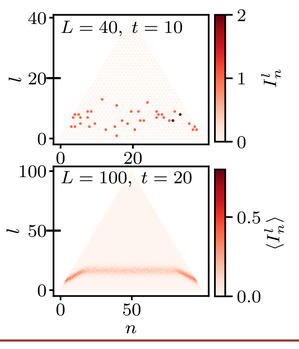
Constraint: every site has exactly two endpoints



Qubits ($q = 2$)



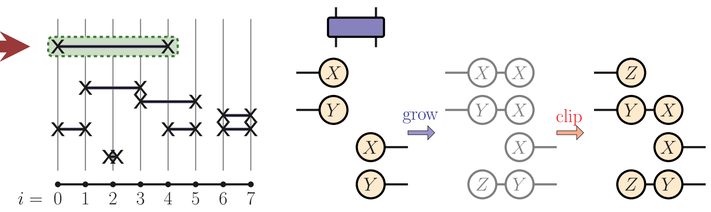
Clifford circuits

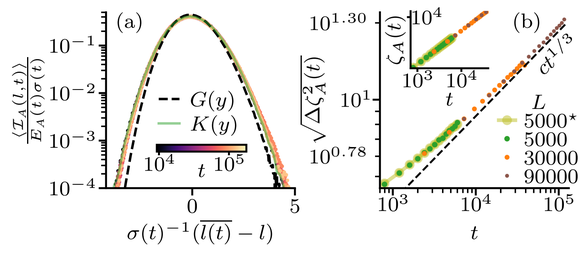


A 'stabilizer state' identified by a set of L mutually commuting and independent Pauli strings $\{\mathcal{Q}_0, \mathcal{Q}_1, ..., \mathcal{Q}_{L-1}\}$ can be simulated efficiently under dynamics generated by the Clifford group.

Bigrams from stabilizer generators

- Leftmost and rightmost sites of support of the stabilizer generators corresponds to a set of bigrams $\{(e^-_\alpha, e^+_\alpha)\}$.
- **Clipped gauge:** Generators can be chosen so that the bigrams satisfy the required constraints.
- Gauge re-imposed after every unitary step.





Phenomenology

"constrained growth of stabilizer generators"

- Local exchange of endpoints biased to increase bigram lengths.
- Shorter bigrams favoured for growth over longer ones.

Outlook

General features

- Rearrangement of local information while preserving the nonlocal constraint.
- Information at shorter length scales scrambles faster.

In presence of noisy dynamics, these features seem lead to $\zeta \sim t$ and $\sqrt{\Delta\zeta^2} \sim t^{1/3}$.

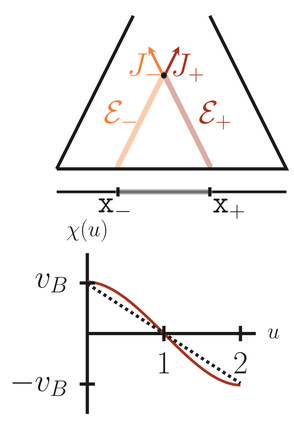
Hydrodynamic theory

$$J_\pm(\mathtt{n},\mathtt{l}) \approx \chi\left(\mathcal{E}_\pm(\mathtt{n},\mathtt{l})\right)\mathtt{I}(\mathtt{n},\mathtt{l})$$

$$\mathcal{E}_\pm = \int_{\mathtt{n}\pm\mathtt{l}/2}^{\mathtt{n}} \mathtt{I}(\mathtt{n}', \mathtt{l} \pm 2(\mathtt{n} - \mathtt{n}'))\, d\mathtt{n}'.$$

For Random circuits,

- $\chi(u) = v_E(1-u)/2$ [4]
- Singular traveling waves $I(\mathtt{n},\mathtt{l}) \approx \delta(\mathtt{l} - 2v_E t)$ in regions unaffected by boundaries.



References

[1] C. Artiaco, T. Klein Kvorning, D. Aceituno Chavez, L. Herviou, and J. H. Bardarson, *Phys. Rev. Lett.* 134, 190401 (2025).

[2] T. Klein Kvorning, L. Herviou, and J. H. Bardarson, *SciPost Phys.* 13, 080 (2022).

[3] A. Nahum, J. Ruhman, S. Vijay, and J. Haah, *Phys. Rev. X* 7, 031016 (2017)

[4] C. Jonay, D. A. Huse, and A. Nahum (2018), arXiv:1803.00089